

TEEP RESEARCH PROJECT B

Cuffless Blood Pressure Estimation

The goal, the two datasets, and how the data was prepared

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Every number here was measured from the data, not quoted from memory

What we are doing, and what we are checking

Blood pressure needs a cuff. A watch already has a sensor that might replace it.

A cuff squeezes your arm to measure blood pressure. It works, but an elderly person living alone cannot manage it five times a day. A smartwatch already shines a light into the skin and records the pulse, all day, for free. That signal is called PPG. Can a computer read blood pressure out of it?

Many papers say yes. But a lot of them tested the wrong way: they put some clips from a person in the training set and other clips from the SAME person in the test set. The model had already met that person. It is like giving a student the exam paper the night before — the marks tell you nothing.

1. How much better does that make the results look?
2. What is the real number, on a person the model has never met?
3. What happens on a real smartwatch instead of hospital equipment?
4. If we give the model a few real readings from the new person, how much does it improve?

Two datasets

One to teach the model. One to check whether it survives outside the hospital.

	PulseDB	PPG-DaLiA
Where it comes from	Hospital ICU and operating rooms	Ordinary daily life
Sensor	Clinical finger sensor + ECG	Smartwatch on the wrist
People	1,553 patients	15 volunteers
Recording per person	1 hour (60 to 100 min, by group)	2.4 hours (88 to 177 min)
Total	1,794 hours · 646,000 clips of 10 s	36 hours
Blood pressure labels	Yes — measured, one per clip	None at all
We use it to	Train the model and test it	Check it on real-world data

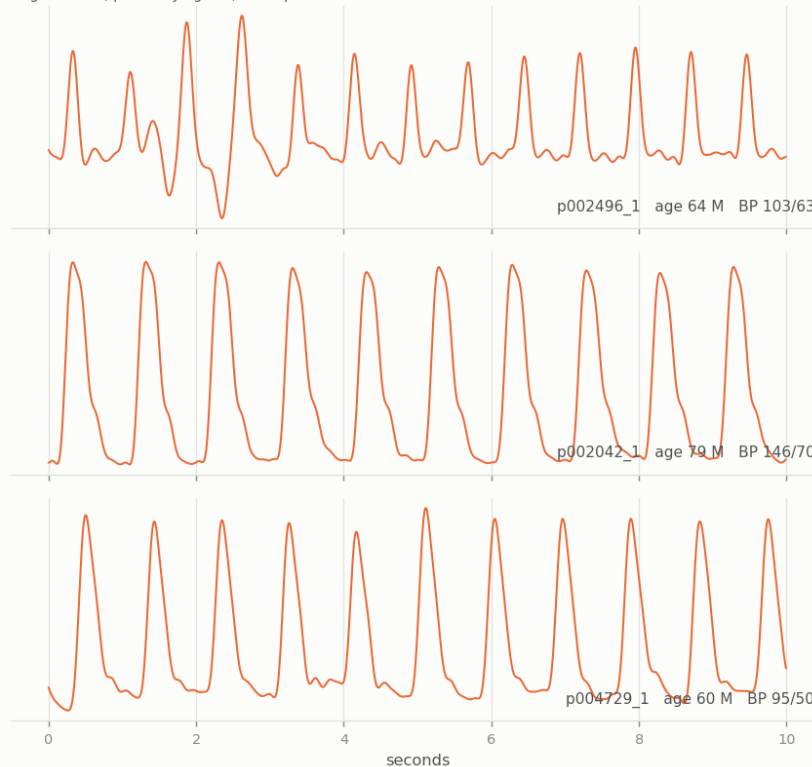
PulseDB is Boston (MIMIC-III) and Seoul (VitalDB) combined, 5,361 patients. The spec says to use the Seoul half only — hence the “VitalDB” filenames. It ships already split into five groups, and every published paper on PulseDB uses those same groups.

Has anyone used both datasets together? One paper has (arXiv:2508.10805, 2025), but for removing motion noise and estimating heart rate — never blood pressure. The pairing exists; using it for blood pressure does not.

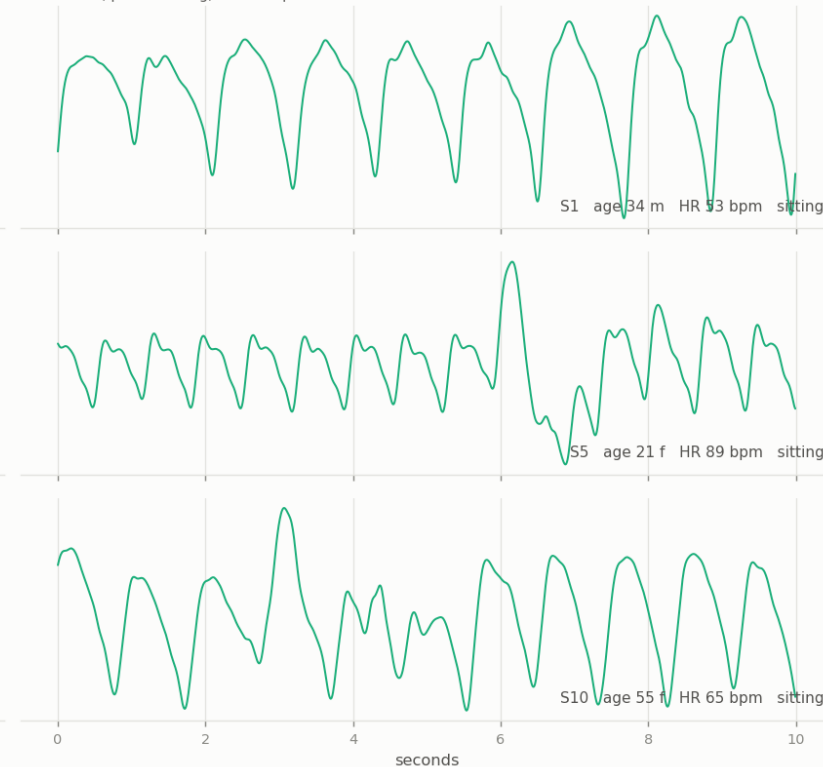
Three real people from each dataset

Same sensor idea, very different conditions.

PulseDB — three hospital patients
finger sensor, patient lying still, blood pressure known



PPG-DaLiA — three volunteers
wrist watch, person sitting, no blood pressure available



Left: hospital patients lying still, blood pressure known. Right: volunteers wearing a watch while sitting, with no blood pressure recorded at all.

Where the five group names came from

They are the dataset's, not ours.

Where the five group names came from

they are the dataset's own official splits, used by every paper on PulseDB

The five groups are defined by PulseDB itself, not by us.

file downloaded		group name we use

VitalDB_Train_Subset.mat	->	train
VitalDB_CalBased_Test_Subset	->	calbased_test
VitalDB_CalFree_Test_Subset	->	calfree_test
VitalDB_AAMI_Cal_Subset	->	aami_cal
VitalDB_AAMI_Test_Subset	->	aami_test

We renamed them to lower case and nothing else. No patient was moved between groups.

WHAT EACH ONE IS FOR

train	the model learns from these patients
calfree_test	'calibration free' - patients the model has never seen and gets no help with. The honest exam.
calbased_test	'calibration based' - the SAME patients as train, different clips. An upper bound, not a real result.
aami_cal	a few clips from 116 patients, used to adapt the model to each one, then scored on the rest.
aami_test	Named after AAMI, the medical device standard.

DO OTHER PAPERS USE THE SAME GROUPS? Yes - all of them.

PulseDB 2023	defined them
Benchmark 2025	Calib / CalibFree / AAMI
DMT 2026	Cal-base / Cal-free / AAMI
UTransBPNet 2024	subject-independent (= calibration free)
rU-Net 2024	calibration-based, plus its own fine-tuning

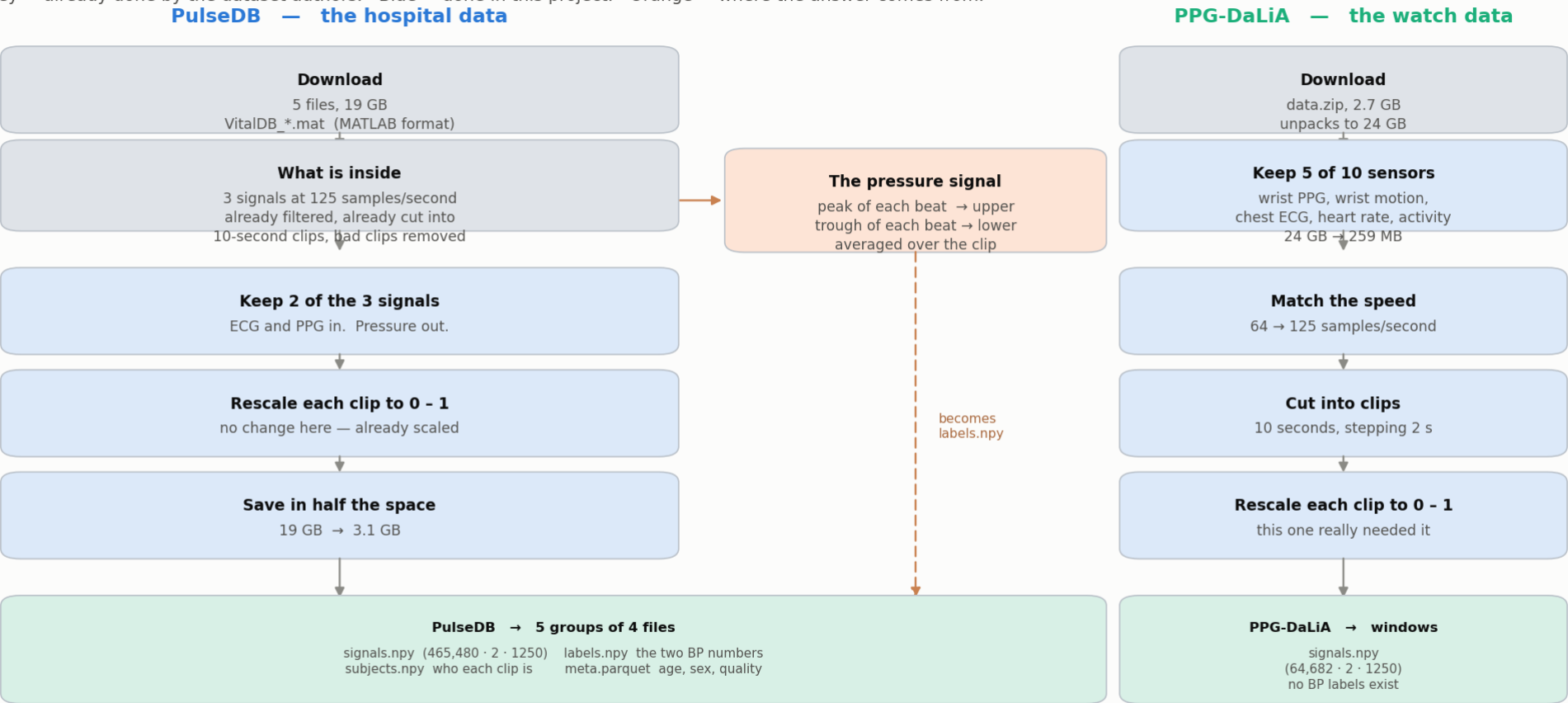
This is why our numbers can be put next to theirs at all.

We lower-cased the file names and changed nothing else. No patient was moved between groups.

The whole pipeline on one page

How the raw downloads become the files a model reads

Grey = already done by the dataset authors. Blue = done in this project. Orange = where the answer comes from.



Both end in exactly the same shape — 2 signals of 1250 numbers per clip — so a model trained on the hospital data runs on the watch data without a single change. That is the only reason the real-world test means anything. Last step, not drawn: patients are split into groups and the code checks that none appears in two of them.

Step 0 — what we actually downloaded

Opening the raw file and printing what is inside it.

Step 0 — what the downloaded file actually contains

read directly from data/raw/pulsedb/, nothing modified

```
>>> f = h5py.File('data/raw/pulsedb/VitalDB_CalFree_Test_Subset.mat')
>>> list(f['Subset'])
```

name	shape	type
Age	(1, 57600)	float64
BMI	(1, 57600)	float64
DBP	(1, 57600)	float64
Gender	(1, 57600)	object
Height	(1, 57600)	float64
SBP	(1, 57600)	float64
Signals	(1250, 3, 57600)	float64
Subject	(1, 57600)	object
Weight	(1, 57600)	float64

HOW TO READ THAT

(1, 57600) = one value per clip, for all 57,600 clips.
Age, SBP, Weight and so on are one number each.

(1250, 3, 57600) = the waveforms.
1250 samples x 3 channels x 57,600 clips

float64 = each number takes 8 bytes. This is why the files are so large.
object = text. Gender is "M"/"F", Subject is an ID like p000053_1.

THE ACTUAL NUMBERS

```
>>> f['Subset']['Signals'][:6, :, 0]
first 6 samples of clip 0, all three channels
```

channel 0 (ECG)	channel 1 (PPG)	channel 2 (ABP)
0.3658	0.2933	68.2310
0.3743	0.2896	74.7662
0.3966	0.2893	82.2520
0.4114	0.2916	89.5596
0.4058	0.2958	96.2278
0.4103	0.3011	101.9762

ECG and PPG sit between 0 and 1 - the dataset authors already rescaled them.

ABP is in real mmHg. Watch it climb 68 -> 102 across these six samples. Six samples is 6/125 second, about 48 milliseconds: this is the heart pumping, caught mid-beat. That rise is what the labels are read from.

SIZE

one clip = 1250 x 3 x 8 bytes = 30,000 bytes
this file = 57,600 clips = 1.7 GB
the training file has 465,480 clips = 13.8 GB

Two things to notice: ECG and PPG arrive already scaled between 0 and 1, and the pressure signal arrives in real units.

Evidence — the file sizes quoted here

Printed off the machine.

Evidence — the file sizes quoted in this report

printed from the machine, not from memory

```
$ ls -lh data/raw/pulsedb/
```

```
13.80 GB VitalDB_Train_Subset.mat
 2.08 GB VitalDB_AAMI_Cal_Subset.mat
 1.71 GB VitalDB_CalFree_Test_Subset.mat
 1.53 GB VitalDB_CalBased_Test_Subset.mat
 0.02 GB VitalDB_AAMI_Test_Subset.mat

19.14 GB total
```

```
$ ls -lh data/processed/pulsedb/train/
```

```
 3.7 MB labels.npy
 5.9 MB meta.parquet
2327.4 MB signals.npy
 29.8 MB subjects.npy
```

```
$ du -sh data/raw data/processed
```

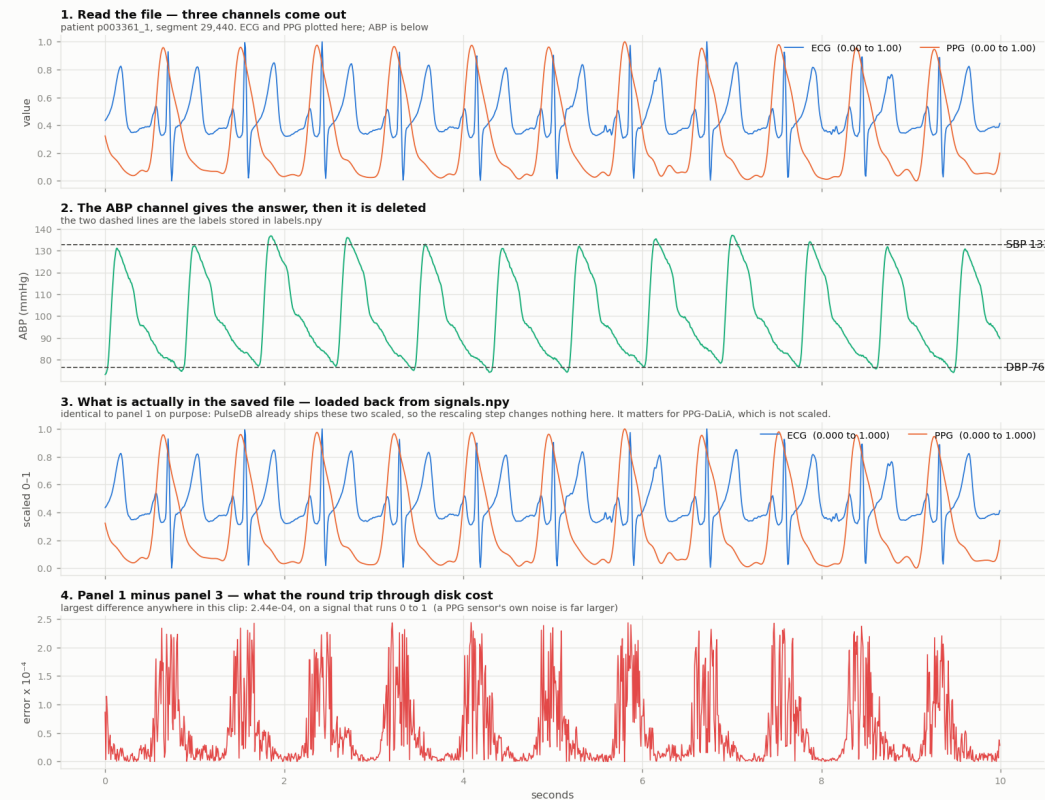
```
20G data/raw the downloads, untouched
3.6G data/processed what a model actually reads
```

The reduction comes from three things: the pressure channel is dropped, each number is stored in half the space, and nothing else changes.

19 GB of downloads become 3.6 GB: the pressure channel is dropped, and each number is stored in half the space.

One real clip, every step

Patient p003361_1, one 10-second clip, from file to stored array.

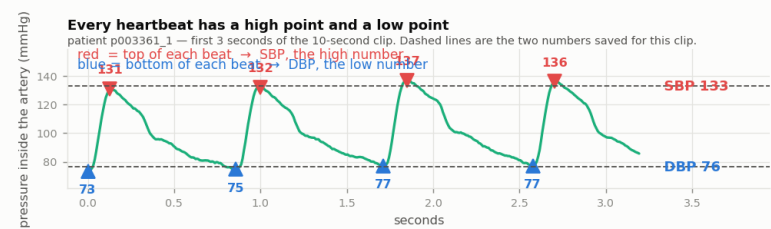


Panel 2 is the important one — the green trace produced this clip's answer of 133 over 76, and is then thrown away. The model only ever sees the blue and orange lines.

Panel 3 looks identical to panel 1 because PulseDB already scales these two channels. That step exists for PPG-DaLiA, which does not.

Where the two numbers come from

A blood pressure reading is the top and the bottom of one wave.



Each red triangle is one heartbeat squeezing.
Each blue triangle is the heart relaxing before the next beat.

That is all blood pressure is: the high number and the low number of the same wave.

OVER THE WHOLE 10-SECOND CLIP

heartbeats found	12
average of the high points	133.4 mmHg
average of the low points	75.6 mmHg

label saved as SBP	132.8 mmHg
label saved as DBP	76.4 mmHg

Our averages land within about 1 mmHg of the saved labels. They are not identical, and that is expected: PulseDB used a published beat-detection algorithm (Elgendi's) to place the peaks, and this figure re-finds them with a simpler method. The labels used everywhere in this project are PulseDB's, not these.

The picture on the left shows only the first 3 seconds, so counting those triangles alone gives a different average again.

A thin tube in the patient's artery measured this directly. Nobody estimated it.

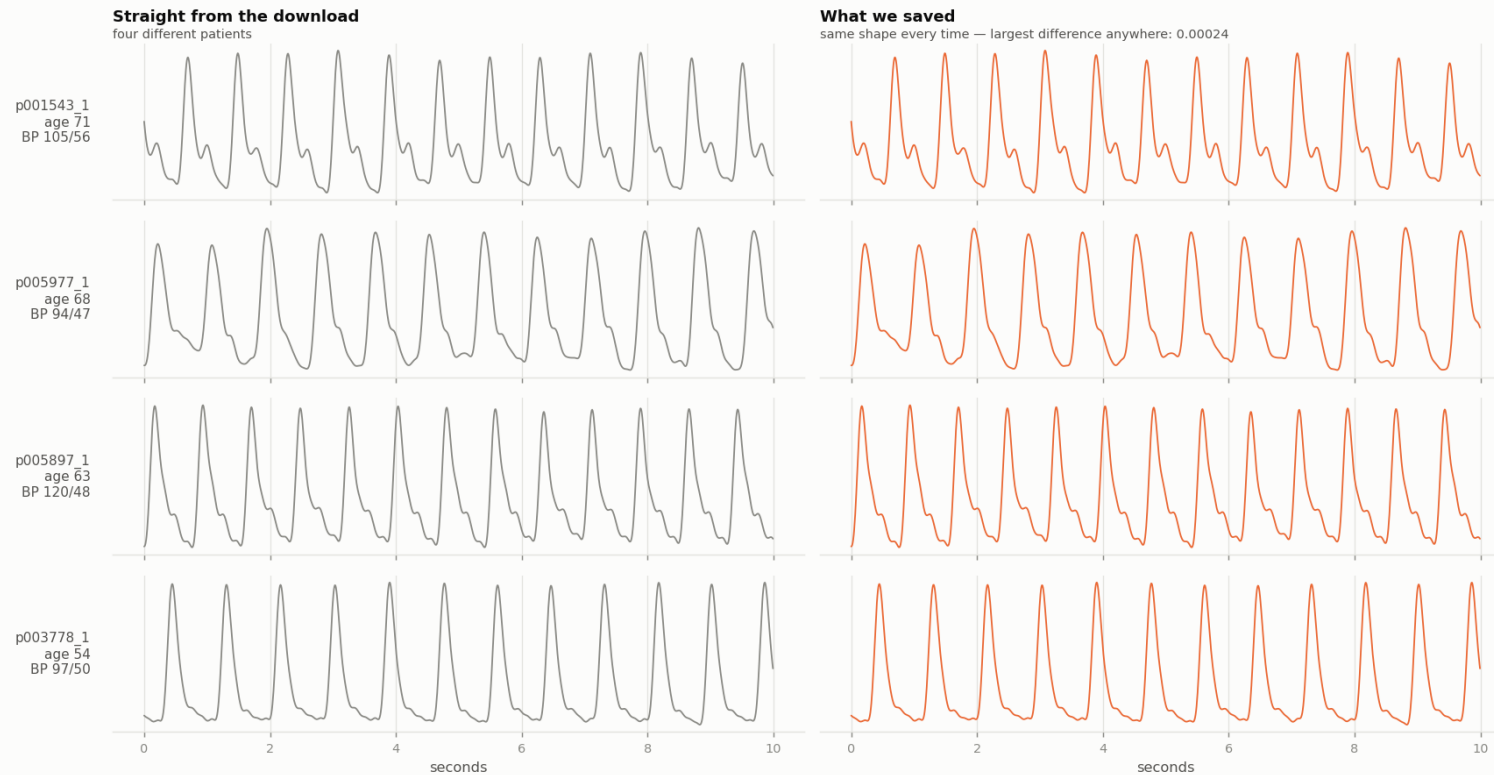
That is why the model is never shown this green line - it would be handing over the answer sheet.

Red triangle = the heart squeezing, the high number. Blue = the heart relaxing, the low number. Average them across the clip and you get 133 over 76.

A thin tube inside the patient's artery measured this. Nobody estimated it — which is why these labels can be trusted, and why the model never sees the green line.

Every patient has a different pulse

Four patients. Left is the download, right is what we saved.



Four people, four clearly different pulse shapes — close to a fingerprint, which is exactly why training and test groups must not share people.

Left and right match to within 0.00024.

Evidence — the authors' own description

PulseDB's published preprocessing, quoted from the paper.

Evidence — the dataset authors' own description

quoted from the published paper

From the PulseDB paper (Wang et al., Frontiers in Digital Health, 2023, doi.org/10.3389/fdgth.2022.1090854), describing what they did before releasing the data:

FILTERING

"a 4th order Chebyshev-II filter at [0.5, 8] Hz before presenting to the Elgendi's algorithm"

BEAT DETECTION

"Pan-Tompkins QRS detection algorithm"

SEGMENTATION

"10-s non-overlapping segments"

QUALITY EXCLUSIONS - a segment was thrown away if it had

"more than 3 consecutive samples of the same value equaling to the minimum or maximum amplitude within the segment, or more than 1 s of the same amplitude"

PPG skewness quality index "< 0"

PPG-to-ABP correlation "< 0.9" after alignment

LABELS

"Beat-to-beat SBP and DBP values were extracted from cycles of the ABP signal ... The reference SBP and DBP values of each segment are thus defined as the average beat-to-beat SBP and DBP values within each segment."

This is why we do not filter the signal again. It is already filtered, already quality-checked, already segmented, already labelled.

Note the last exclusion rule: they compared PPG against the pressure trace "after alignment" - so the authors knew these two channels are not recorded on the same clock.

Note the last exclusion rule: they compared PPG against the pressure trace “after alignment”, so the authors knew the two channels do not share a clock.

The two datasets arrive in very different states

PulseDB comes finished. PPG-DaLiA does not — which is why it needed more work.

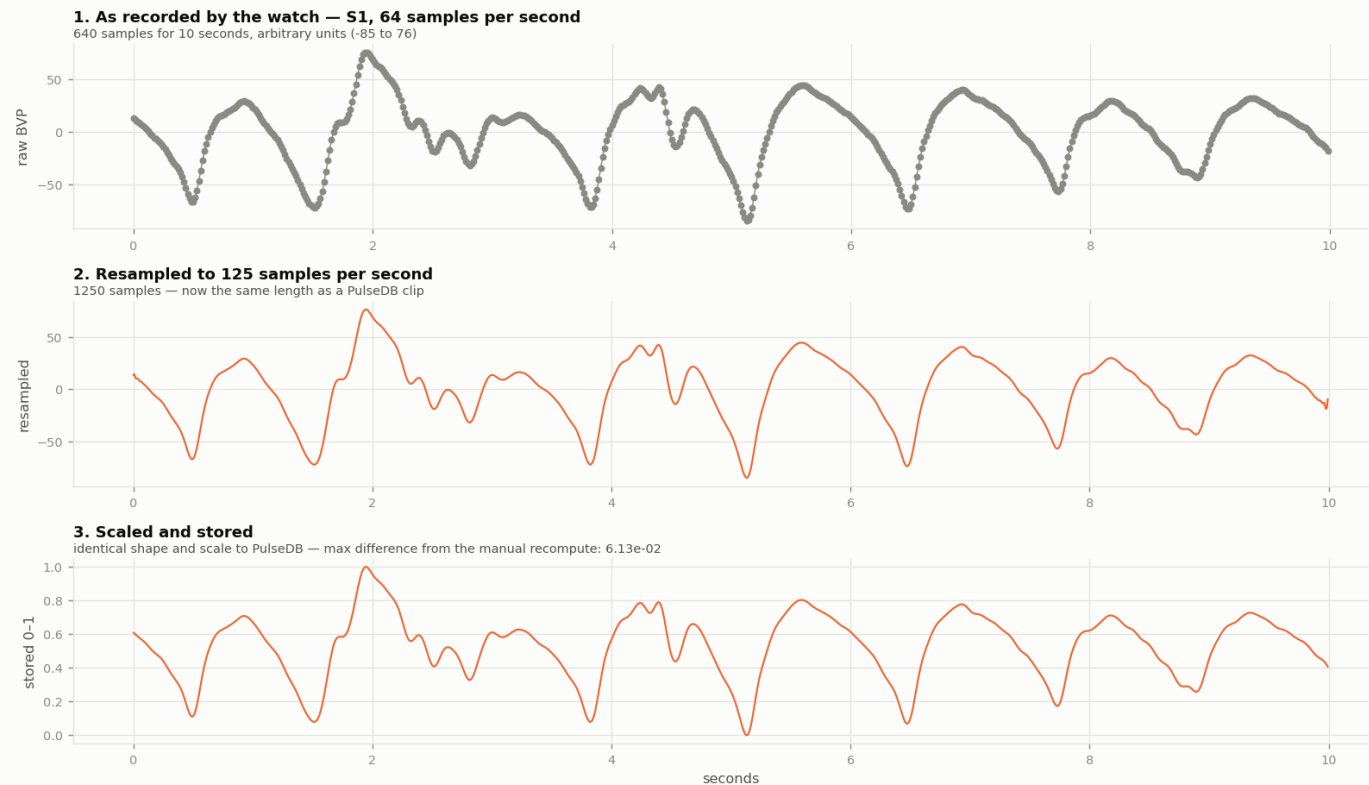
Step	PulseDB	PPG-DaLiA	So who did it?
Sensors synchronised	done by authors	done by authors	neither of us
Signal filtered	done — Chebyshev-II, 0.5–8 Hz	not done	nobody — see note
Cut into fixed clips	done — 10 s, non-overlapping	not done	we did, for DaLiA
Bad clips removed	done — three quality rules	not done	we score quality, keep all
Signal rescaled	done — 0 to 1	not done	we did, for DaLiA
Sampling rate	125 Hz	64 Hz wrist, 700 Hz chest	we resampled DaLiA to 125
Blood pressure labels	yes — from an arterial line	none at all	cannot be created

PulseDB's authors shipped the data already filtered, segmented, quality-checked and labelled, so we left it exactly as it came. Filtering an already filtered signal removes real detail rather than cleaning it.

PPG-DaLiA's watch does its own processing on the device and the authors ship what it produced, so we did not add a filter there either — the real-world test only means something if both are treated identically. Its readme suggests 8-second clips; we use 10 so the shape matches PulseDB.

PPG-DaLiA, the same clip through each step

Volunteer S1, ten seconds from ten minutes into the recording.



The watch records 640 numbers for these ten seconds; PulseDB expects 1250. Resampling fills the difference without changing the pulse shape.

Evidence — what was inside the PPG-DaLiA download

24 GB of sensors, of which this project uses five.

Evidence — what the PPG-DaLiA download contained

and what of it this project actually uses

```
$ unzip -l data/raw/ppg_dalia/data.zip
```

downloaded archive	2.86 GB compressed
other	24.14 GB uncompressed
subject.pkl (synced)	20.28 GB uncompressed
RespiBAN.h5 (raw chest)	3.82 GB uncompressed
E4 raw wrist	0.04 GB uncompressed
TOTAL	48.27 GB

WHAT WE KEPT

wrist PPG	64 Hz	the model input
wrist motion	32 Hz	tells us when the arm was moving
chest ECG	700 Hz	reference heartbeat
heart rate	0.5 Hz	label, derived from the chest ECG
activity	4 Hz	label, what the person was doing

WHAT WE DROPPED

chest accelerometer, EMG (muscle), EDA (sweat), temperature, respiration
- none of them are used anywhere in this project

result: 48.3 GB -> 259 MB across 15 files

The archive is kept, so any dropped channel can be recovered later with one command.

Most of it is the chest sensor suite from the stress-detection side of the study. The archive is kept, so any dropped channel can be recovered.

What we ended up with

Group	Clips	Patients	Per patient	What it is for	Size
train	465,480	1,293	60 min	The model learns from this	2.33 GB
calfree_test	57,600	144	67 min	The real exam — brand new patients	288 MB
aami_cal	70,212	116	101 min	A few readings to tune the model	351 MB
aami_test	666	116	1 min	Then score it on those same people	3 MB
calbased_test	51,720	1,293	7 min	Easy case — patients it knows	259 MB
PPG-DaLiA	64,682	15	144 min	Watch data, no BP labels	323 MB

Each group is four files: the waveforms, the two BP numbers, which patient each clip came from, and a small table of age, sex, weight and signal quality. They line up row by row.

One thing not to misread: PPG-DaLiA's clips are cut every 2 seconds and are 10 seconds long, so they overlap. Its recording really is 36 hours in total, not 180.

What one finished group looks like

The training group, file by file.

File	What is in it	Size
signals.npy	The waveforms — 465,480 clips, each 2 signals x 1250 numbers	2,327 MB
labels.npy	The answers — 465,480 rows, an upper and a lower BP number	3.7 MB
subjects.npy	Which patient each clip came from — 465,480 names	29.8 MB
meta.parquet	Age, sex, height, weight, BMI and three signal-quality scores	5.9 MB

clip 0 belongs to patient p000001_1, a 48-year-old man, blood pressure 131 / 88,
waveform 2 signals of 1250 numbers each.

The waveform file is opened without loading it — a training run reads only the clips it is about to use, so a 2.3 GB file costs no memory to open.

Three things that could have gone wrong

Each one was tested, not assumed.

What could go wrong	How we checked	What came back
The saved file might not match the original	Reloaded four clips from the download and compared them number by number	Same to within 0.00024 — that is 0.02%
Saving in a smaller format might have damaged the signal	Ran pulse detection on both versions and compared where it found the heartbeats	Identical on all 300 clips tested
A patient used for training might turn up in a test group	Compared the patient lists of every pair of groups	Zero shared, and it is re-checked on every run

The third one matters most. If a training patient ever appeared in a test group, the results would quietly get better and nobody would notice — so the build stops instead.

Two groups do share patients on purpose (aami_cal with aami_test, train with calbased_test). Both are by design, and neither is ever a headline result.

Did we prepare the data the same way as the papers?

Our settings against the two most relevant studies, and how much data each had.

Setting	PulseDB gives	Benchmark 2025	DMT 2026	Us	Same?
Measurements per second	125	125	125	125	Yes
Clip length	10 s / 1250	10 s / 1250	1250	10 s / 1250	Yes
Cleaning / filtering	already done	kept as is	not stated	kept as is	Yes
Patient groups	3 official ones	official	official	official	Yes
Input signals	ECG + PPG	PPG only	PPG only	ECG + PPG	No
Rescaling	0 to 1	not stated	z-score	0 to 1	No
Training clips	—	418,986	465,480	465,480	Yes

Four of six settings match exactly, because they come with the dataset and nobody can change them. Our preprocessing is the standard preprocessing for this dataset, not something we invented.

The two that differ are deliberate: we keep the ECG because the spec asks for it, and we use the 0-to-1 scaling the dataset already provides rather than adding another step.

The last row matters: the best published result trained on 465,480 clips. We hold exactly the same number.

Six papers, and what each was trying to do

Paper	Year	Its goal
PulseDB	2023	Build a large, clean benchmark with patient IDs and standard test rules
Validation study	2024	Check whether deep models hold up on a large benchmark
rU-Net	2024	Push accuracy as high as possible, then adapt to each new patient
Benchmark study	2025	Compare five architectures fairly, then test them on outside data
DMT	2026	Improve accuracy by feeding the model the patient's age, sex and BMI
This project	2026	Measure how much the test rule inflates results, then fix it with personalization

Their numbers differ partly because they were answering different questions, not because one method is better than another.

Models used, and results on seen vs new patients

Paper	Model	Patients it had seen	Brand new patients
PulseDB	1D ResNet-18	not reported as a final number	not reported as a final number
Validation study	CNN-LSTM-attention, UTransBPNet	not reported separately	12.50 / 8.32 (best variant)
rU-Net	U-Net + ResNet + attention	4.49 / 2.69	4.14 / 2.48 after tuning on 10% of the new patient
Benchmark study	LeNet1D, XResNet1d50/101, Inception1D, S4	9.08 / 6.08	12.70 / 8.05
DMT	Transformer + FiLM demographic conditioning	4.56 / 2.62	12.17 / 7.89
This project	Gradient boosting, Random Forest (CNN / ResNet / Transformer next)	7.39 / 4.38	12.93 / 8.24

Every paper that reports both shows the same pattern: around 4 to 9 mmHg on patients it had already seen, around 12 to 14 on new ones.

That gap is the honest difficulty of the problem, and it is what this project set out to measure.

Is there a paper using BOTH our datasets?

Yes — one. The distinction matters for what we can claim.

	Reduction of motion artifacts from PPG signals using...
Published	Physiological Measurement / arXiv:2508.10805, 2025
Its task	Removing motion artifacts, then estimating heart rate
Uses PulseDB for	training — clean clips with synthetic motion artifacts added
Uses PPG-DaLiA for	testing — real motion from daily life
Blood pressure?	Never predicts or evaluates blood pressure

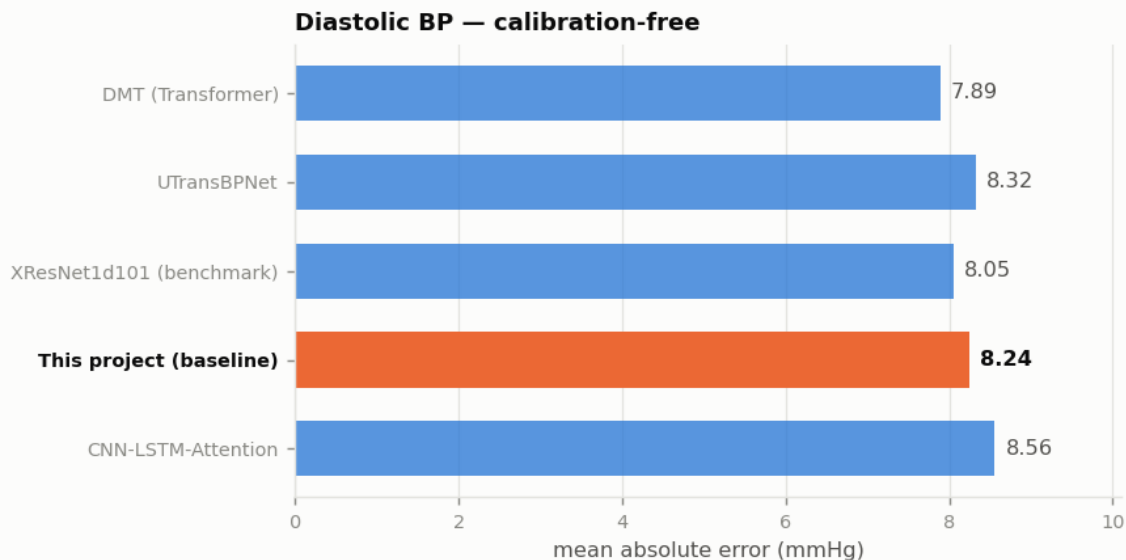
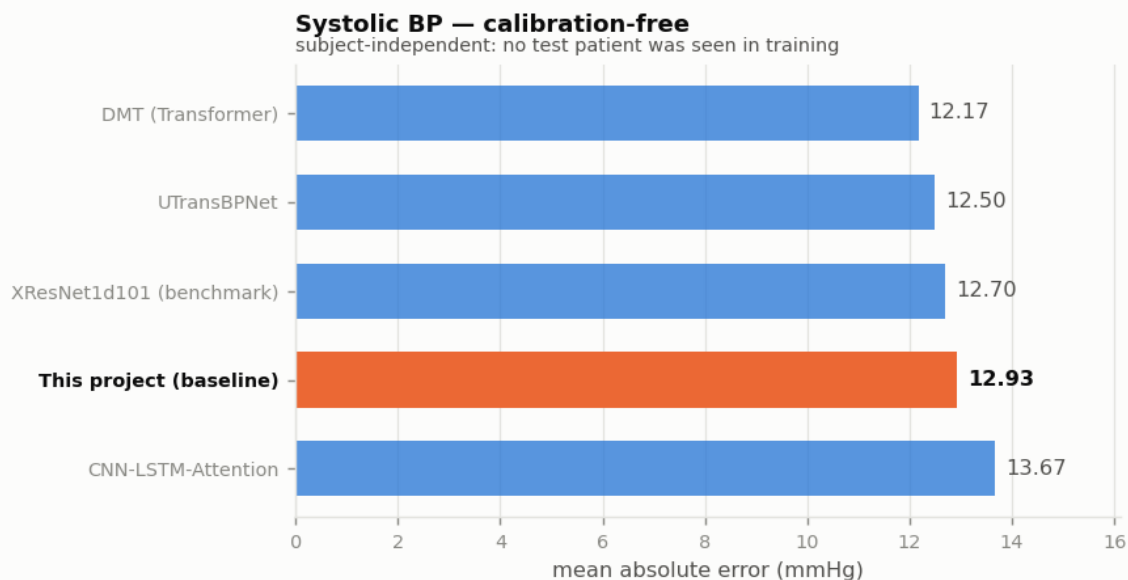
So the pairing has been done — but for cleaning up noisy signals and estimating heart rate, not for blood pressure.

Our claim has to be precise: nobody has trained a blood-pressure model on PulseDB and tested it on PPG-DaLiA. Saying “nobody has used these two together” would be wrong, and a reviewer would find this.

Our result against the published ones

Tested on brand new patients. Lower is better.

Published results on PulseDB, honest protocol — orange is this project

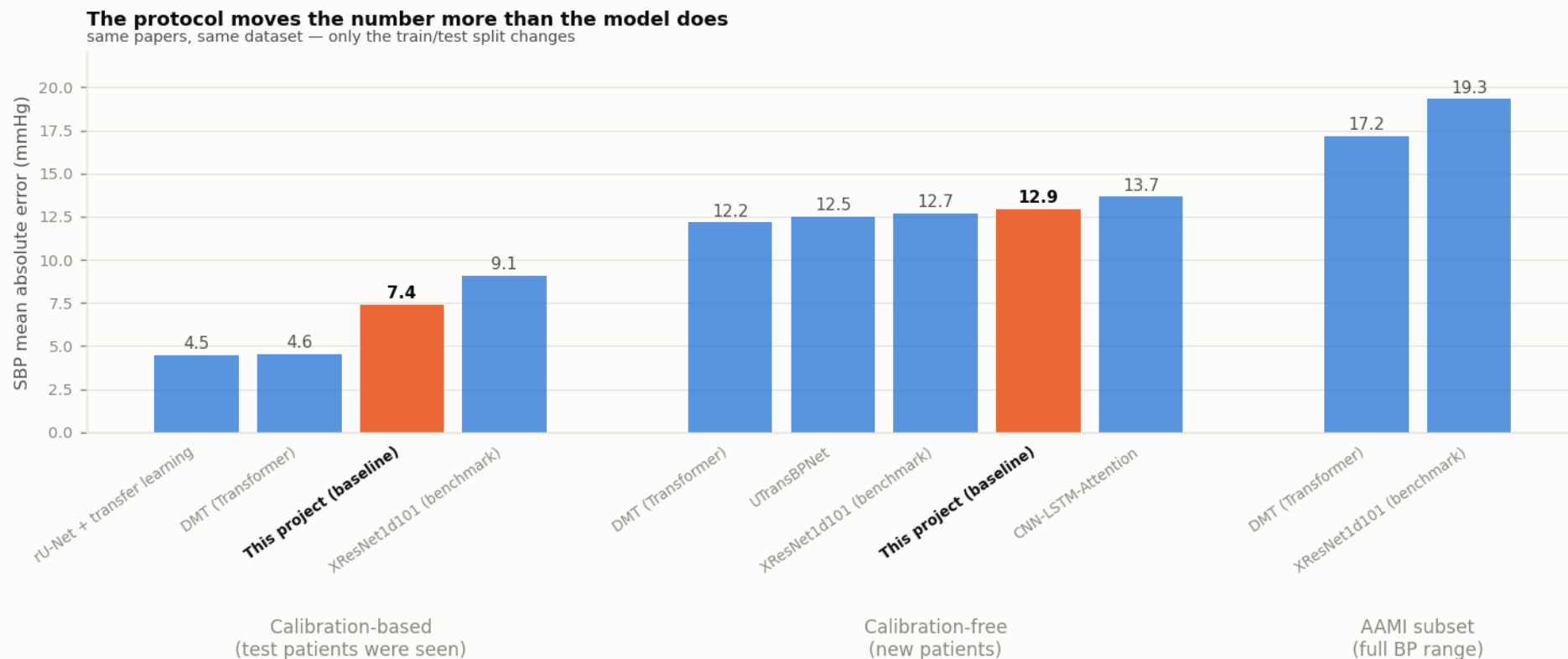


Ours is the orange bar — gradient boosting on 60 hand-measured features, with no tuning.

0.76 mmHg behind the best Transformer. Four years of new architectures have moved this number by 1.5 mmHg in total.

The biggest factor is not the model

Same models, same data — only which patients are in the test set changes.



4.5 mmHg when the test patients had been seen in training. 12.9 when they are new. 19.3 on the hardest test set.

What we have actually trained so far

Being precise, because the word “training” is easy to overstate.

Model	Type	Status	Fits run
HistGradientBoosting	Gradient boosting	Trained	18
Random Forest	Decision trees	Trained	6
1D CNN	Deep learning	Code ready, not run	0
ResNet1D	Deep learning	Code ready, not run	0
Transformer	Deep learning	Code ready, runs on Colab	0

24 models fitted so far. Each run trains 6 — one for the upper and one for the lower BP number, across the three ways of splitting the patients.

No deep model has been trained yet.

References

PulseDB — Wang et al., Frontiers in Digital Health, 2023. doi.org/10.3389/fdgth.2022.1090854

PPG-DaLiA — Reiss et al., UCI Machine Learning Repository, dataset 495.

Generalizable deep learning for PPG-based BP estimation — a benchmarking study, 2025. [arXiv:2502.19167](https://arxiv.org/abs/2502.19167)

DMT: Demographic Conditioning, Morphology-Enhanced Transformer, 2026. [arXiv:2606.11125](https://arxiv.org/abs/2606.11125)

Validation of deep learning models for cuffless BP estimation, 2024.

rU-Net, Multi-Scale Feature Fusion and Transfer Learning, IEEE JBHI, 2024.

Investigation of Data Leakage in Deep-Learning-Based Blood Pressure Estimation, IEEE, 2023.

Reduction of motion artifacts from PPG signals using learned convolutional sparse coding, 2025.
[arXiv:2508.10805](https://arxiv.org/abs/2508.10805)